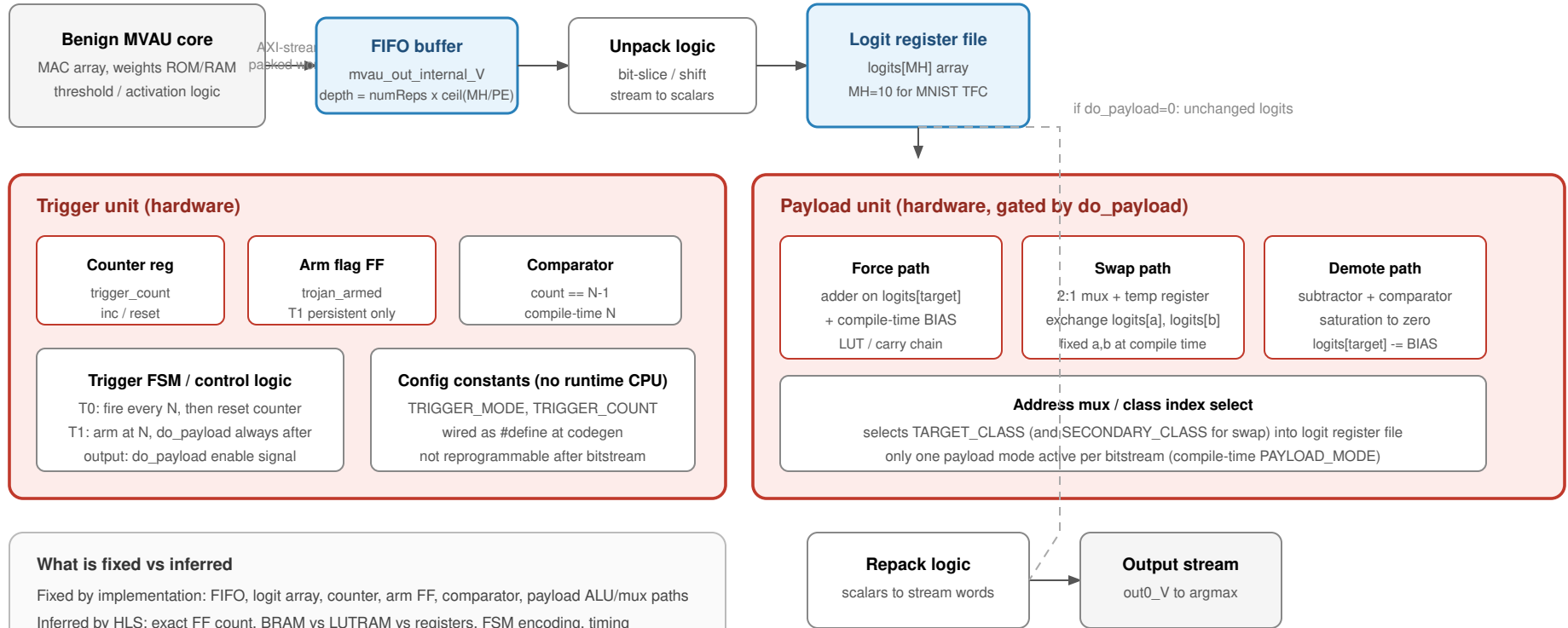


Hardware Components in the Final-MVAU Trojan (Post-HLS View)

Logical structure from generated HLS; exact resource mapping depends on Vitis HLS synthesis



What is fixed vs inferred

Fixed by implementation: FIFO, logit array, counter, arm FF, comparator, payload ALU/mux paths

Inferred by HLS: exact FF count, BRAM vs LUTRAM vs registers, FSM encoding, timing

Verify after synthesis: Vivado utilization / HLS report on final MVAU IP

TFC MNIST: MH=10 logits, small register file; trojan area is tiny vs full MVAU MAC array

Legend

Storage (FIFO / register file - typically BRAM, SRL, or FFs)

Trojan control + datapath (FFs, comparators, add/sub, muxes)

Standard FINN MVAU compute (dominant area)

All trojan parameters (N, mode, payload, target class, bias) are compile-time constants in the bitstream, not software-configurable at runtime.